

Problem: Constellation Twin Matching Game

After solving the ruin's triangle puzzle, Rhythm hears a rumble. A hidden passage opens. Dhruva waits inside pointing to four constellations namely, Orion, Canis Major, Taurus and Auriga. Each of them have nine stars indexed from 1 to 9.

Dhruva claims 18 stars, making them shine brightly. The remaining 18 become dim and now belong to Rhythm.

The game starts lasting a total of 18 rounds. In each round, Dhruva announces one of his bright stars. Rhythm must respond with one of his dim stars. Rhythm wins the round iff he responds with a twin star (i.e. it is in the same constellation or has the same index as Dhruva's star). Stars are discarded after each round once they are announced.

What is the maximum number of rounds Rhythm is guaranteed to win, no matter what Dhruva selects?